

INTRODUCTION TO DSP

ENSC 429 – Digital Signal Processing

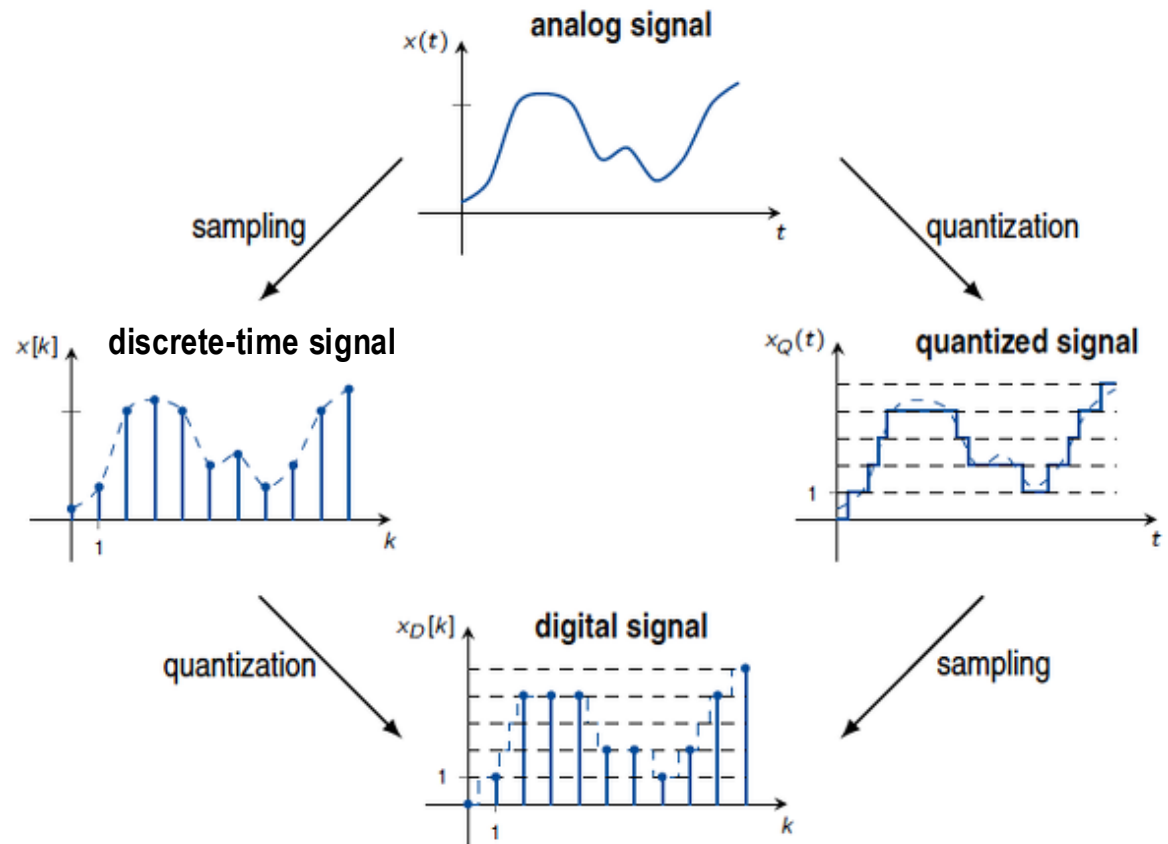
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WHAT IS DSP?

Digital Signal Processing

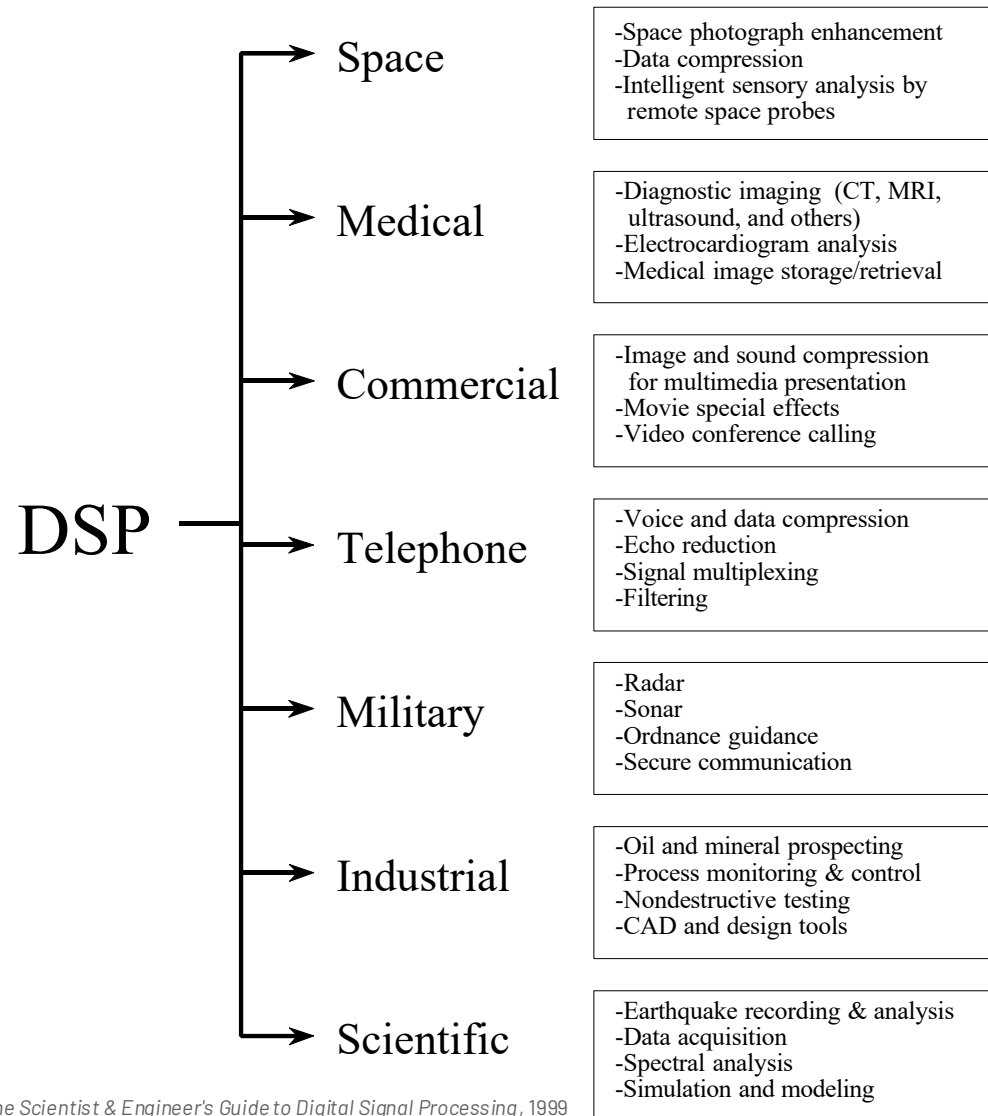
- **Digital**: discrete time (space) and discrete amplitude
- **Signal**: any measurement, or function
- **Processing**: (deliberate) manipulation, to achieve a goal



From <https://www.javatpoint.com/difference-between-analog-signals-and-digital-signals>

APPLICATIONS OF DSP

- DSP is all around you
- You use it every day when you
 - Use your phone / tablet / computer
 - Watch TV
 - Listen to music
 - Play video games
 - Use AR/VR
 - Drive
 - ...



From S. W. Smith, *The Scientist & Engineer's Guide to Digital Signal Processing*, 1999

SELECTED APPLICATIONS

A phone call (underlined steps indicate where DSP is used)

- The phone rings (ringtone is a digital audio file)
- You pick up and say: *“Hello”*
 - Your voice is captured, sampled, digitized, and compressed using a speech codec
 - Bits from the compressed voice bitstream are modulated and transmitted from your phone via an antenna
- Antenna picks up the response
 - The response signal is demodulated, decompressed, and played out through your phone’s speaker: *“I’m sending you a photo”*
 - The next signal is demodulated, decompressed (it was a JPEG file), and displayed on your phone’s screen
- ...
- You get the picture...

SELECTED APPLICATIONS

Watching TV (underlined steps indicate where DSP is used)

- You enter a room and say “*Hey Google, turn on the TV*”
 - Your voice is captured, sampled, digitized, and passed through a speech recognition engine to fulfill your command
- The TV turns on, news is playing
 - The screen displays a composite video with news-anchor on the left, images of space on the right, and TSX ticker on the bottom
 - The anchor’s voice says: “*Scientists have found a new potentially habitable planet. Using the latest advances in radio astronomy, synthetic aperture imaging, and interferometry, the new planet was discovered in a solar system 64 light years away. We are showing you the picture captured by the James Webb telescope.*”
- ...
- You get the picture...

SELECTED APPLICATIONS

Driving (underlined steps indicate where DSP is used)

- You enter your car and turn on the engine
 - The engine is quiet because of the active noise control
- You shift to reverse to pull out of the parking spot
 - The video from the rear camera is displayed on the control panel
 - As you reverse, the radar on the side of our car warns you of another vehicle coming behind you
- You turn on the radio
 - But you don't like the music they're playing, so you connect your phone via Bluetooth and play your favorite Spotify playlist (MP3 files)
- You exit the parking lot
 - The camera at the exit recognizes your license plate and charges your account
- ...
- You get the picture...

HISTORY

Foundational knowledge related to DSP goes back a long time...

- Newton used finite-difference methods to approximate solutions of differential equations
 - Such difference equations are common models for discrete-time systems
- In 1805, Gauss used efficient computations with complex exponentials, which form the basis of the Fast Fourier Transform (FFT), even before Fourier's work on harmonic analysis
- In the 1940's, work on sampled-data systems (e.g., radar and early digital control systems) provides motivation for developing the theory behind discrete-time processing
- In the 1950's, seismic signals recorded on magnetic tape were digitized and analyzed on a computer
- Also in the 1950's, speech coding was being investigated in an effort to digitize the backbone of the telephone network

HISTORY

- 1965: Fast Fourier Transform (FFT) by Cooley and Tukey
- Also in the 1960's and 1970's, early work on digital image processing
- In the 1980's, the work on digital video processing and compression leads to the first digital video coding standards: H.120 and H.261
- Also in the 1980's: Compact Disc (CD)
- 1992:
 - JPEG (Joint Photographic Experts Group) image coding standard
 - MPEG-1 (Moving Picture Experts Group) video coding standard
 - MPEG-1 Audio Layer 3 (MP3) audio coding standard
- Many commercial applications followed...

READING ASSIGNMENT

- Read Chapter 1 in the textbook
- Start reading Chapter 2